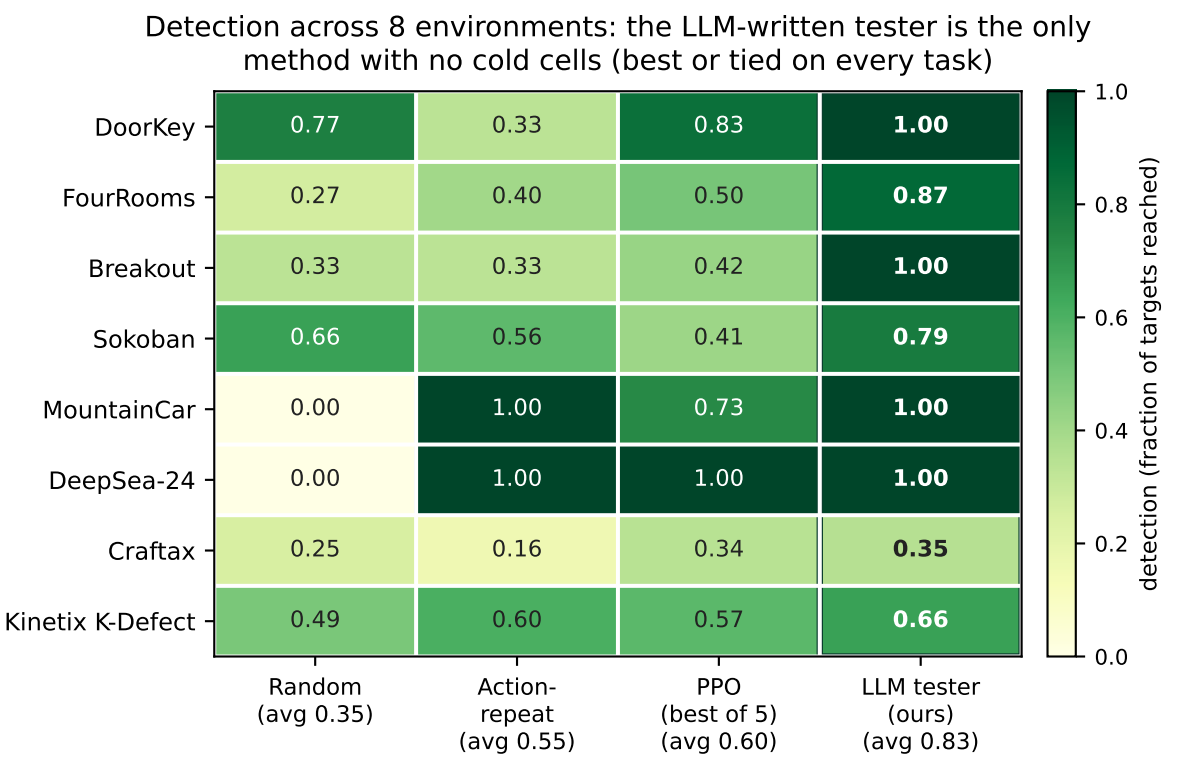
Detection across 8 environments: the LLM-written tester is the only method with no cold cells (best or tied on every task)
DoorKey
FourRooms
Breakout
Sokoban
MountainCar
DeepSea-24
Craftax
Kinetix K-Defect
0.77
0.33
0.83
1.00
0.27
0.40
0.50
0.87
0.33
0.33
0.42
1.00
0.66
0.56
0.41
0.79
0.00
1.00
0.73
1.00
0.00
1.00
1.00
1.00
0.25
0.16
0.34
0.35
0.49
0.60
0.57
0.66
Random (avg 0.35)
Action-repeat (avg 0.55)
PPO (best of 5) (avg 0.60)
LLM tester (ours) (avg 0.83)
detection (fraction of targets reached)
1.0
0.8
0.6
0.4
0.2
0.0